PAPER_233: SGR 1745-2900 Enhanced — SMBH Tidal Proximity Coupling, Static B-Field Magnetic Energy, ATNF Pulse Period

Author: Daniel T. Murphy

SGR 1745-2900 is the closest known magnetar to a supermassive black hole, located at a projected separation of 0.09–0.3 pc from Sagittarius A and a deprojected estimate of* ~ 0.92 *pc. This paper documents three MUGE terms absent from the Session 53 implementation (*`MagnetarSGR1745DynamicModulationCalculator`*): (1) SMBH tidal coupling acceleration* $a_{BH} = G M_{SgrA*}/r_{BH}^2$*, (2) static (non-decaying) magnetic stored energy* $a_{mag} = B^2 V/(2\mu_0 Mr)$ *with* $B = 2 \times 10^{10}$ *T, and (3) use of the ATNF-catalogued pulse period* $P = 3.76$ *s (replacing an inferred value). The superconductive suppression factor is also refined to* $f_{sc} = 1 - B/B_{crit}$*.*

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Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grok*share*8d951e12.txt extraction — Doc 14 enhanced) **Date:** March 2026 **Classification:** Enhanced MUGE — Three New Terms vs Session 53 MagnetarSGR1745DynamicModulationCalculator **Status:** Proof-Quality Whitepaper

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Abstract

SGR 1745-2900 is the closest known magnetar to a supermassive black hole, located at a projected separation of 0.09–0.3 pc from Sagittarius A* and a deprojected estimate of ~ 0.92 pc. This paper documents three MUGE terms absent from the Session 53 implementation (`MagnetarSGR1745DynamicModulationCalculator`): (1) SMBH tidal coupling acceleration $a_{BH} = G M_{SgrA*}/r_{BH}^2$, (2) static (non-decaying) magnetic stored energy $a_{mag} = B^2 V/(2\mu_0 Mr)$ with $B = 2 \times 10^{10}$ T, and (3) use of the ATNF-catalogued pulse period $P = 3.76$ s (replacing an inferred value). The superconductive suppression factor is also refined to $f_{sc} = 1 - B/B_{crit}$.

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1. Physical System

SGR 1745-2900's proximity to Sgr A* makes it unique in the magnetar population:

Parameter	Value
Position	~ 0.09 pc projected, ~ 0.92 pc deprojected from Sgr A*
M	$1.4 M_\odot$ (NS)
r	20 km (NS radius)
B	2×10^{10} T (static; not decaying)
B_{crit}	4.4×10^{13} T
P_{init}	3.76 s (ATNF Pulsar Catalogue)
τ_Ω	8000 yr
L_0	5×10^{28} W

Parameter	Value
τ_d	1000 s
M_{SgrA*}	$4 \times 10^6 M_\odot$
r_{BH}	0.92 pc = 2.83×10^{16} m

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2. New Terms vs Session 53

Comparison Table

Term	Session 53	Session 58 Enhancement
BH proximity	Absent	$a_{BH} = GM_{SgrA*}/r_{BH}^2$
B field	Decaying $B(t) = B_0 e^{-t/\tau_B}$	Static $B = 2 \times 10^{10}$ T
Superconductivity	Generic f_{TRZ}	$f_{sc} = 1 - B/B_{crit}$
Pulse period	Inferred from P_{dot}	$P = 3.76$ s (ATNF)
Magnetic energy	Absent	$a_{mag} = B^2 V/(2\mu_0 Mr)$
Burst decay	Absent	$a_{decay} = L_0\tau_d(1-e^{-t/\tau_d})/(Mr)$

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3. SMBH Tidal Coupling (Novel in SGR 1745 context)

$$a_{BH} = \frac{G M_{SgrA*}}{r_{BH}^2} = \frac{6.674 \times 10^{-11} \times 4 \times 10^6 \times 1.989 \times 10^{30}}{(2.83 \times 10^{16})^2}$$

$$a_{BH} = \frac{5.309 \times 10^{26}}{8.01 \times 10^{32}} \approx 6.63 \times 10^{-7} \text{ m/s}^2$$

This tidal coupling from the $4 \times 10^6 M_\odot$ SMBH is **dominant over the magnetar's self-gravity** at 0.92 pc, where the self-gravity $GM_{NS}/r_{NS}^2 \approx 2 \times 10^{12}$ m/s^2 only at the neutron star surface.

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4. Static Magnetic Energy (Novel vs Session 53)

$$a_{mag} = \frac{B^2}{2\mu_0} \cdot \frac{V_{NS}}{Mr} = \frac{(2 \times 10^{10})^2}{2 \times 4\pi \times 10^{-7}} \cdot \frac{(4\pi/3)(20000)^3}{2.785 \times 10^{30} \times 20000}$$

Session 53 used a decaying $B(t)$; this implementation uses a **static** B field, reflecting evidence that SGR 1745-2900's field has remained stable since its 2013 activation.

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5. Superconductive Factor

$$f_{sc} = 1 - \frac{B}{B_{crit}} = 1 - \frac{2 \times 10^{10}}{4.4 \times 10^{13}} = 1 - 4.55 \times 10^{-4} \approx 0.99955$$

Applied to the base gravity: $a_{base} = GM/r^2 \cdot (1 + H_0 t) \cdot f_{sc}$ — a ~0.05% suppression from near-critical field.

6. ATNF Pulse Period

The pulse period `P = 3.76` s is directly from the ATNF Pulsar Catalogue (2024), more precise than the `P \approx 3.8` s sometimes quoted. This affects the spin-down back-reaction:

```
\frac{d\Omega}{dt} = -\frac{2\pi}{P \tau_\Omega} e^{-t/\tau_\Omega}
```

7. Calculator Class

```
class SGR1745BHProximityMagEnergyCalculator(_CP3Calculator):
"""PAPER_233: SGR 1745-2900 enhanced — BH tidal a_BH, static mag energy, ATNF P=3.76s"""
# Session 58 — grok_share_8d951e12.txt Doc 14 enhanced
```

Located in `CondensedPhysics3.py` (Session 58).

8. Conclusion

The enhanced SGR 1745-2900 MUGE adds three physically motivated terms absent from Session 53: SMBH tidal coupling (dominant at 0.92 pc), static magnetic energy density, and ATNF-calibrated pulse period. The result is the most complete MUGE treatment of any Galactic Centre magnetar in the UQFF library.

Source: grok*share*8d951e12.txt — Doc 14 (SGR 1745-2900 BH Proximity Enhanced MUGE)

UQFF computed: Eddington luminosity UQFF correction = 1 - [SSq]■exp(-?■?t) = 1 - 5.7e-1 ■ exp(-2.9e-4) = 4.3e-1; F_U at event horizon = 2.0e+18 m/s■.